ICE 417 Assignment I – Solved

Question 1

Minimize

$$Z = 20x_1 + 10x_2$$

Subject to:

$$x_1 + 2x_2 \leq 40 \; 3x_1 + x_2 \quad \geq 30 \; 4x_1 + 3x_2 \geq 60 \; x_1, x_2 \quad \geq 0$$

Solution (Graphical Method)

The feasible region is obtained by plotting all constraints and identifying the intersection points that satisfy all inequalities.

The feasible corner point that satisfies all constraints is:

$$(x_1, x_2) = (6, 12)$$

Evaluating the objective function:

$$Z = 20(6) + 10(12) = 240$$

Optimal Solution

$$\boxed{x_1 = 6,\ x_2 = 12,\ Z_{\min} = 240}$$

Question 2

An oil-producing firm manufactures products **X** and **Y**.

Decision Variables

- x = tonnes of product X per day
- y = tonnes of product Y per day

LP Formulation

Maximize

$$Z = 80x + 120y$$

Subject to:

$$x + y \leq 9 \; 20x + 50y \quad \leq 360 \; x \geq 2 \; y \quad \geq 3 \; x, y \geq 0$$

Solution (Graphical Method)

Solving the intersection of the binding constraints:

$$x + y = 9$$

$$20x + 50y = 360$$

From substitution:

$$y = 6, \; x = 3$$

Profit:

$$Z = 80(3) + 120(6) = 960$$

Optimal Solution

$$\boxed{x = 3, \; y = 6, \; Z_{\max} = 960}$$

Question 3

A wire-producing company manufactures wire types **w_1** and **w_2**.

Decision Variables

- w_1 = units of wire type 1 per week
- w_2 = units of wire type 2 per week

LP Formulation

Maximize

$$Z = 60w_1 + 40w_2$$

Subject to:

$$w_1 \leq 25 \; w_2 \quad \leq 35 \; 2w_1 + w_2 \leq 60 \; w_1, w_2 \quad \geq 0$$

Solution (Graphical Method)

Evaluating corner points of the feasible region gives the maximum value at:

$$(w_1, w_2) = (12.5, 35)$$

$$Z = 60(12.5) + 40(35) = 2150$$

Optimal Solution

$$\boxed{w_1 = 12.5,\ w_2 = 35,\ Z_{\max} = 2150}$$

Sensitivity (Objective Function)

The solution remains optimal as long as the ratio of profit coefficients remains within the slope bounds of the binding constraints.

Question 4

A manufacturer produces models **X** and **Y**.

Decision Variables

- x = units of model X
- y = units of model Y

LP Formulation

Maximize

$$Z = 50x + 30y$$

Subject to:

$$2x + y \geq 18\ x + y \quad \geq 12\ 3x + 2y \leq 34\ x, y \quad \geq 0$$

Solution

Solving the intersection of active constraints gives:

$$x = 8,\ y = 4$$

$$Z = 50(8) + 30(4) = 520$$

Optimal Solution

$$\boxed{x = 8,\ y = 4,\ Z_{\max} = 520}$$

Question 5

Maximize

$$Z = 2x_1 + x_2$$

Subject to:

$$x_1 + 2x_2 \leq 10\ x_1 + x_2 \quad \leq 6\ x_1 - x_2 \leq 2\ x_1 - 2x_2 \quad \leq 1\ x_1, x_2 \geq 0$$

Solution (Graphical Method)

Evaluating feasible corner points shows the maximum occurs at:

$$(x_1, x_2) = (4, 2)$$

$$Z = 2(4) + 2 = 10$$

Optimal Solution

$$\boxed{x_1 = 4,\ x_2 = 2,\ Z_{\max} = 10}$$

End of Assignment